

Bren Alden

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EDUCATION

Bina Nusantara University – Tangerang, Banten

September 2023 - Present

School of Computer Science - Bachelor of Computer Science (Artificial Intelligence)

GPA: 3.82 / 4.00 (6th Semester)

Scholarship: Student Advisory and Support Center Mentor Scholarship Awardee

Relevant Coursework: Artificial Intelligence; Machine Learning; Deep Learning; Natural Language Processing; Speech Recognition; Computer Vision; Software Engineering; Database Technology; Object Oriented Programming

PROFESSIONAL EXPERIENCE

PT. Astra Daihatsu Motor – North Jakarta, Jakarta

March 2026 – Present

Robotic Process Automation Developer Intern (Python & Power Automate)

- Developed and delivered 15+ end-to-end RPA projects using Python and Microsoft Power Automate, covering User Requirement gathering and UAT, and reducing average manual work hours by 90%.
- Resolved major RPA bugs to improve processing stability, reduce execution time and manual workload, and maintain laptop performance during automation runs; developed web-based monitoring and validation tools to support RPA operations across ADM.
- Led 5 RPA Workshops within a 3-month internship, training ADM employees in Python-based automation for Email, Excel, SAP, and Website processes.

PROJECTS

RAVE (RPA Authentication and Validation Engine) – Company's Project

- Developed a centralized RPA monitoring web portal built with Next.js, TypeScript, and Tailwind CSS to track bot execution workflows, performance, and operational status in real time.
- Integrated Qwen2.5-Coder LLM for automated code analysis, detecting security vulnerabilities, logic errors, and potential execution risks in RPA automation scripts prior to deployment.
- Fine-tuned a 7.6B parameter Qwen LLM using Unsloth and PEFT (LoRA), training only 0.53% of parameters to adapt the model for domain-specific RPA code vulnerability detection.

PadiCure (AI-based Rice Leaf Disease Detection) – Competition's Project – [Repositories](#)

- Developed and designed a web-based rice disease detection platform with an integrated image scanner for identifying potential rice diseases.
- Implemented a transfer learning approach using a pre-trained ResNet50 convolutional neural network, adapting its architecture for multi-class rice leaf disease classification through custom training and classification layers.
- Achieved a validation accuracy of 90.54%, demonstrating the model's capability to effectively classify rice leaf diseases across the validation dataset.

WorkAIEvaluation (AI-based Workout Evaluator) – Coursework's Project – [Repositories](#)

- Developed a computer vision system using MediaPipe pose extraction with CNN-LSTM models to analyze exercise form, count valid repetitions, and score workout sets (push-up, pull-up, squat, plank) in real time.
- Achieved low validation loss (~0.03–0.07 MSE) across four exercise-specific models through iterative architecture tuning and learning-rate scheduling, indicating high consistency between predicted and actual form/ROM measurements.

PUBLICATIONS

Analyzing Deep Learning in Quick, Draw!: A Case Study on Hand-drawn Sketch Recognition

August 2025

ICITCOM 2025 – [DOI](#) – [Certificate](#)

- Investigated alternative deep learning models (ViT-S/16 and ResNet-50) to surpass the traditional CNN used in Google Quick, Draw! for hand-drawn sketch recognition, achieving up to 94.60% accuracy across category sizes ranging from 10 to 50.
- Benchmarked model performance based on accuracy, loss, parameter count, and inference time, finding ViT-S/16 more efficient in size while ResNet-50 offered greater stability and speed as category complexity increased.

SKILLS

- Programming Languages:** Python, C, Java, JavaScript, SQL, HTML, CSS, R
- Developer Tools:** VSCode, Google Colab, Power Automate, Git, Power BI
- AI Concepts:** LLM, CNN, ANN-SNN, LSTM, Forecasting, Prompt Engineering, Recommendation System
- AI Libraries/Framework:** LangChain, NLTK, Scikit-learn, TensorFlow, Torch, Keras, MediaPipe, Unsloth, Matplotlib
- RPA Libraries/Framework:** Selenium, Playwright, Openpyxl, Pywin32, BeautifulSoup, PyAutoGUI
- Soft Skills:** Leadership, Teamwork, Public Speaking, Mentoring, Teaching, Project Management
- Languages:** Native Indonesian Speaker; Fluent in English; Elementary in Mandarin